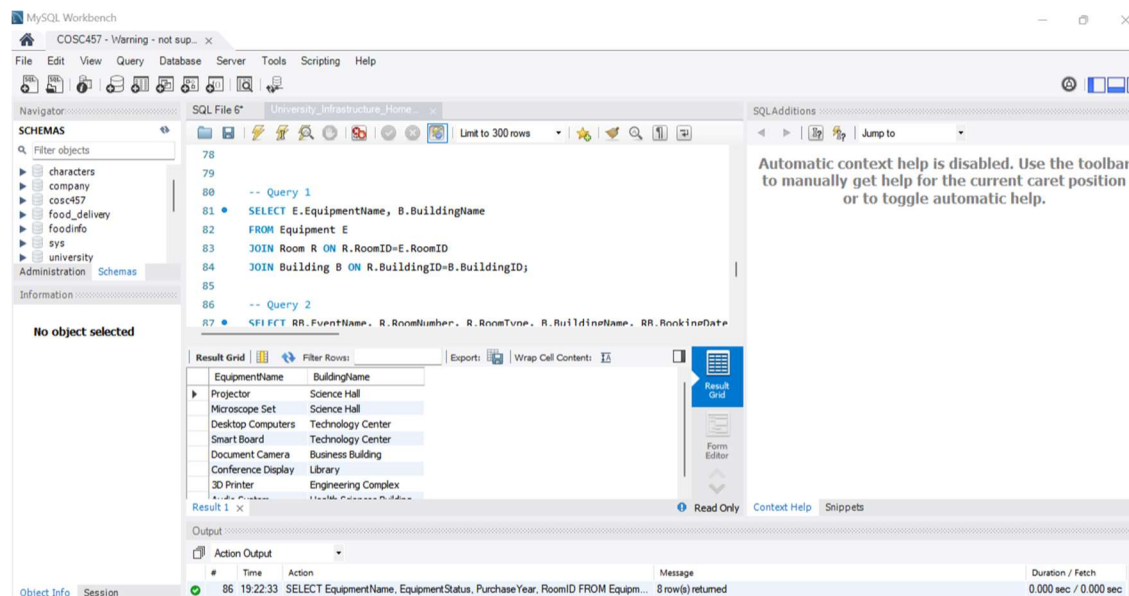


Advanced SQL HW

Query 1: The university wants to identify which equipment belongs to which building. Explain which tables must be connected to answer this question and identify the foreign key relationships involved. Write a query for this.

Building, Equipment, and Room must be connected to answer this question since you need to know about building and equipment relationships, and equipment is assigned to a room located within a building. The foreign keys involved in this will be RoomID and BuildingID.

```
SELECT E.EquipmentName, B.BuildingName
FROM Equipment E
JOIN Room R ON R.RoomID=E.RoomID
JOIN Building B ON R.BuildingID=B.BuildingID;
```



Query 2: Write a query to show each booked event along with the room number, room type, building name, booking date, and number of attendees. Only show rooms that have actual bookings.

```
SELECT RB.EventName, R.RoomNumber, R.RoomType, B.BuildingName,
RB.BookingDate, RB.NumberOfAttendees
FROM RoomBooking AS RB
JOIN Room R ON R.RoomID=RB.RoomID
JOIN Building B ON B.BuildingID=R.BuildingID;
```

The screenshot shows the MySQL Workbench interface. The SQL Editor contains the following SQL code:

```

84 JOIN Building B ON R.BuildingID=B.BuildingID;
85
86 -- Query 2
87 SELECT RB.EventName, R.RoomNumber, R.RoomType, B.BuildingName, RB.BookingDate
88 FROM RoomBooking AS RB
89 JOIN Room R ON R.RoomID=RB.RoomID
90 JOIN Building B ON B.BuildingID=R.BuildingID;
91
92 -- Query 3
93 SELECT RoomNumber, RoomType, Capacity, BuildingID

```

The Result Grid displays the following data:

EventName	RoomNumber	RoomType	BuildingName	BookingDate
Intro Biology Lecture	SH-101	Classroom	Science Hall	2026-02-03
Python Programming Lab	TC-115	Computer Lab	Technology Center	2026-02-04
University Research Talk	TC-220	Lecture Hall	Technology Center	2026-02-05
Business Analytics Workshop	BB-140	Classroom	Business Building	2026-02-06
Graduate Study Group	LIB-300	Study Room	Library	2026-02-07
Chemistry Review Session	SH-101	Classroom	Science Hall	2026-02-08

The Output pane shows the execution of Query 2, returning 8 rows.

Query 3: Write a query to find rooms whose capacity is greater than the average capacity of all rooms in the university. Display the room number, room type, capacity, and building ID. Write this query as a nested query.

```

SELECT RoomNumber, RoomType, Capacity, BuildingID
FROM Room
WHERE Capacity > (
    SELECT AVG(Capacity)
    FROM Room
);

```

The screenshot shows the MySQL Workbench interface. The SQL Editor contains the following SQL code:

```

90 JOIN Building B ON B.BuildingID=R.BuildingID;
91
92 -- Query 3
93 SELECT RoomNumber, RoomType, Capacity, BuildingID
94 FROM Room
95 WHERE Capacity > (
96     SELECT AVG(Capacity)
97     FROM Room
98 );
99

```

The Result Grid displays the following data:

RoomNumber	RoomType	Capacity	BuildingID
SH-101	Classroom	80	1
TC-220	Lecture Hall	120	2
BB-140	Classroom	60	3

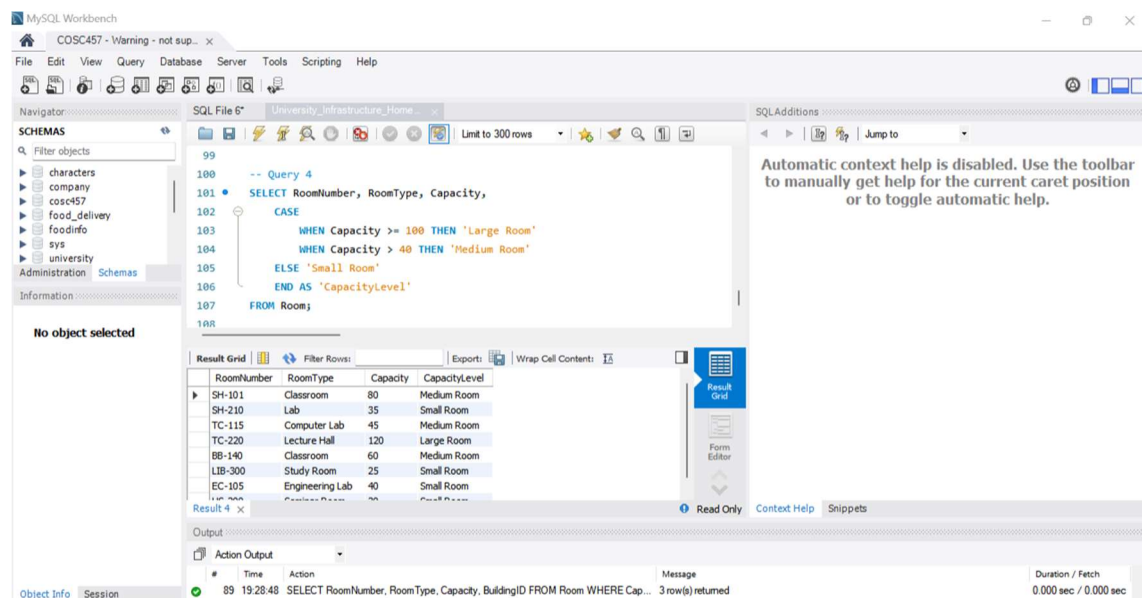
The Output pane shows the execution of Query 3, returning 3 rows.

Query 4: Write a query to label each room based on its capacity:

- Capacity 100 or more = Large Room
- Capacity between 40 and 99 = Medium Room
- Capacity below 40 = Small Room

Display the room number, room type, capacity, and the capacity label.

```
SELECT RoomNumber, RoomType, Capacity,
       CASE
           WHEN Capacity >= 100 THEN 'Large Room'
           WHEN Capacity > 40 THEN 'Medium Room'
           ELSE 'Small Room'
       END AS 'CapacityLevel'
FROM Room;
```



Query 5: Write a query to display all equipment ordered first by equipment status and then by purchase year, with the oldest equipment appearing first within each status category.

Display equipment name, equipment status, purchase year, and room ID.

```
SELECT EquipmentName, EquipmentStatus, PurchaseYear, RoomID
FROM Equipment
```

```
ORDER BY EquipmentStatus, PurchaseYear;
```

The screenshot shows the MySQL Workbench interface. The SQL Editor contains the following query:

```
104     WHEN Capacity > 40 THEN 'Medium Room'
105     ELSE 'Small Room'
106     END AS 'CapacityLevel'
107 FROM Room;
108
109 -- Query 5
110 • SELECT EquipmentName, EquipmentStatus, PurchaseYear, RoomID
111     FROM Equipment
112     ORDER BY EquipmentStatus, PurchaseYear;
```

The Results window displays the following data:

EquipmentName	EquipmentStatus	PurchaseYear	RoomID
Conference Display	Needs Repair	2016	106
Desktop Computers	Needs Repair	2018	103
Audio System	Under Maintenance	2015	108
Document Camera	Working	2017	105
Projector	Working	2019	101
Smart Board	Working	2020	104
Microscope Set	Working	2021	102

The Output window shows the execution of the query:

#	Time	Action	Message	Duration / Fetch
90	19:29:11	SELECT RoomNumber, RoomType, Capacity, CASE WHEN Capacity >= 100 THE...	8 row(s) returned	0.000 sec / 0.000 sec